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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester MCA (CBCS) Degree Examinations
May – June 2025

22MCA213 – SOFT COMPUTING

Duration: 3 Hours

Max. Marks: 100

Note: Answer any Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the following membership function with an example for each i) Singleton Membership Function ii) Gaussian Membership Function iii) Triangle Membership Function iv) Trapezoidal Membership Function	10	L*2	1	1
b)	Explain the following Defuzzification methods with an example for each. i) Lambda-cut method ii) Last of Maxima iii) Mean of Maxima iv) First of Maxima	10	L2	1	1
2. a)	Define fuzzy logic with a suitable example and explain its advantages and disadvantages over classical logic.	10	L2	1	1
b)	What is a fuzzy rule? Explain fuzzy IF-THEN rules with suitable real-world examples.	05	L2	1	1
c)	(b) A fuzzy set for temperature is given as: $T = \{(15^\circ\text{C}, 0.1), (20^\circ\text{C}, 0.4), (25^\circ\text{C}, 0.8), (30^\circ\text{C}, 1), (35^\circ\text{C}, 0.6), (40^\circ\text{C}, 0.3)\}$ Determine the defuzzified value using the weighted average method.	05	L3	1	2

Unit – II

3. a) Let A and B be the fuzzy numbers,

 $A(x) =$

$$\begin{cases} 0 & 3 < x \leq -1 \\ (x+1)/2 & -1 < x \leq 1 \\ (3-x)/2 & 1 < x \leq 3 \end{cases}$$

 $B(x) =$

$$\begin{cases} 0 & 5 < x \leq 1 \\ (x-1)/2 & 1 < x \leq 3 \\ (5-x)/2 & 3 < x \leq 5 \end{cases}$$

Perform addition and subtraction arithmetic operations on the given fuzzy number.

10 L3 2 2

P.T.O.

b)	Define fuzzy proposition. Briefly explain the four types of fuzzy propositions.	10	L2	2	1
4. a)	Explain the different properties a fuzzy set must have to qualify as a fuzzy number with an example.	10	L2	2	1
b)	Fuzzy sets A and B are defined as: $A = \{ (2, 0.9), (3, 0.5), (4, 1) \}$ $B = \{ (2, 0.8), (3, 1), (4, 0.8) \}$ Evaluate the following operations: i) Complement of union ii) A intersection B iii) Cartesian product of A and B iv) Algebraic sum v) Bounded sum	10	L3	2	2

Unit – III

5. a)	What is a perceptron? Explain its architecture and working mechanism with a suitable diagram and derive the perceptron training rule.	10	L2	3	1
b)	Describe the Backpropagation Algorithm. Provide a step-by-step explanation along with its mathematical derivation.	10	L2	3	1
6. a)	Explain the architecture of an Artificial Neural Network (ANN). Describe its different layers—input layer, hidden layers, and output layer—and their roles in processing information. Discuss the types of neural network architectures, including feedforward and recurrent networks, with suitable examples.	10	L2	3	1
b)	Explain the learning paradigms of Artificial Neural Networks (ANNs). Discuss Supervised Learning, Unsupervised Learning, and Reinforcement Learning with suitable examples for each.	10	L2	3	

Unit – IV

7. a)	Describe the architecture of a Neuro-Fuzzy System with a neat diagram. Explain the roles of fuzzification and defuzzification in the system.	10	L2	4	1
b)	Explain the role of Fuzzy Logic in economics, and pattern recognition, providing relevant examples for each.	10	L2	4	1
8. a)	Explain the concept of Neuro-Fuzzy Systems. Discuss how they integrate neural networks and fuzzy logic. Write any five applications.	10	L2	4	1
b)	Discuss the application of Fuzzy Logic in medical diagnosis. Explain how it is utilized in disease recognition and decision-making, providing relevant examples.	10	L3	4	1

Unit – V

9. a)	Describe the implementation of a Genetic Algorithm for solving the Traveling Salesman Problem (TSP).	10	L2	5	1
b)	Explain the five phases of a Genetic Algorithm in detail. Illustrate the process with a suitable flowchart.	10	L2	5	1
10. a)	Explain different types of mutation operators used in Genetic Algorithms. Provide an example for each type to illustrate their role in maintaining genetic diversity.	10	L2	5	1
b)	Discuss Genetic Algorithms (GA). Explain their concept, working principles, and significance in solving optimization problems.	10	L3	5	1